

PRAHAR MULTI-DETECTOR AUDIT ARCHITECTURE WHITEPAPER

Algorithmic Sentinel & Continuous Integrity Framework for Statutory Fund Flows

1. Executive Summary

PRAHAR is an AI-powered automated audit sentinel engineered for the Members of Parliament Local Area Development Scheme (MPLADS). It provides autonomous, non-partisan, multi-tier anomaly detection across project lifecycles, financial disbursements, vendor bidding behaviors, and geographical execution realities.

2. The Seven Specialized Detection Engines

Engine ID	Detector Name	Methodology / Principle	Primary Target Anomaly
DET-01	Benford First-Digit Law	Logarithmic distribution analysis of sanction amounts	Manufactured & manipulated tender figures
DET-02	Split Tender Sentinel	Detection of sub-threshold clusters below statutory limits	Bypassing formal e-procurement thresholds
DET-03	Vendor Cartel & Network	Bipartite graph clustering & bid-rotation detection	Collusive vendor cartels & shared bank accounts
DET-04	Geographical Co-Location	Haversine clustering & GIS boundary overlap	Ghost works & duplicate geo-tagged physical works
DET-05	DSR Schedule Discrepancy	Automated parsing against state Schedule of Rates	Inflated itemized unit rates & billing padding
DET-06	Milestone Velocity Anomaly	Z-score timeline analysis vs historical district norms	Premature invoice clearance before physical completion
DET-07	Demographic Quota Sentinel	Census SC/ST boundary intersection	Statutory non-compliance with 15% SC / 7.5% ST

3. Risk Scoring & Statutory Reporting

PRAHAR synthesizes the individual detector scores into a calibrated Composite Anomaly Score (0 to 100). Scores ≥ 70 trigger critical alert status, notifying the MoSPI Central Nodal Agency and the District Vigilance Committee. All findings are exportable into cryptographically signed PDF and CSV statutory audit digests.